

Rudra Nath Sinha

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OBJECTIVE

A results-driven and detail-oriented Chemical Engineering student with hands-on experience in industrial training and research projects. Seeking to leverage analytical and problem-solving skills in a challenging role within a dynamic organization to contribute to innovative solutions and optimize processes.

EDUCATION

Course	College / University	Percentage	Year
B.Tech(Chemical Engineering)	Birla Institute of Technology, Mesra Ranchi	85.8 (Current)	2021-Present
ISC (Class XII)	De Nonili School C.M.R.I. Dhanbad	87.5%	2021
ISCE (Class X)	De Nonili School C.M.R.I. Dhanbad	87.1%	2019

EXPERIENCE

- IVL Dhunseri Petrochem Industries Private Limited** 1ST May to 31st May
Core Equipment and Technological Advancements in Polymer Production Summer Training
 - **Production Technology & Industry Overview**
 - * **Key Reactors:**Central PTA and MEG driving esterification, polycondensation and polymerisation reactors .
 - * **Polymerization:** PTA & MEG achieved 15% increase in efficiency in polycondensation & Disk Ring reactor.
 - * **Distillation:**Upgraded columns can ensure 52 %saperation efficeincy of MEG and water.
 - **Project: Optimization Industrial Distillation Processes: A Detailed Design of Water-Ethylene Glycol Column**
 - * **Efficiency Optimization:**Enhanced design features improved reactor output from 23% to 52%
 - * **Distillation Unit Design:**optimised Reflux Ratio to enhance purity and separation
 - * **Residence Time Optimization:**Reduced the Residence time fo liquid holdup on tray by 40sec
 - **Project: Design and Analysis of a Shell and Tube Heat Exchanger for Industrial Applications**
 - * **Heat Transfer Rate:**Achieved 150 kW/m², ensuring effective thermal exchange within one month.
 - * **Heat Exchange Area:**Designed with a total heat exchange area of 25m²
 - * **Residence Time Optimization:**Reduced the Residence time fo liquid holdup on tray by 40sec
- TATA STEEL** 5Th June to 3st July
Coke Oven Lifecycle Enhancement through Research Summer Training
 - **Production Technology & Industry Overview**
 - * **Coke Oven Batteries:** Examined coke oven batteries 8,9,10&11, each producing 5.25 million tons annually.
 - * **Product Varieties:**Analyzed over 25 steel grades, and their applications.
 - * **By-Product Plant Analysis:**Gained in-depth knowledge of by-product generation (including sulphar, Tar and naphthalene) processes identifying resource recovery and cost reduction.
 - **Project: Establishing gas pressure measurement system of Carbolite oven to improve coke oven's life cycle**
 - * **Data Analysis:**Analyzed pressure data to identify trends and improve operational stability.
 - * **Gas Pressure Measurement:**Developed a gas pressure measurement system relating 25 Kg oven & 7 Kg oven
 - * **Outcome Impact:**71.5% of the variance in gas pressure can be explained by the polynomial model.
- TATA STEEL** July 2023 to 12th August 2023
Data Visualisation: Empowering Business with Effective Insights Virtual Experience Program
 - **Business Scenario:**Framed the business scenario to provide context and background.
 - **Choosing Visuals:**Selected the most appropriate visuals for effective representation.
 - **Creating & Communicating:**Created effective visuals and communicated insights clearly.
- BANK OF AMERICA** July 2023 to August 22nd, 2023
Investment Banking Virtual Experience Program Virtual Experience Program
 - **Performance & SWOT:**Analyzed 5 years of financial data and conducted a SWOT analysis.
 - **Strategic Alternatives & Modeling:**Explored 3 strategic options and created 2 financial models.
 - **Communication & Relationships:**Applied 4 communication strategies & developed relationship building approach
- JPMorgan Chase** July 2023 to 12th August 2023
Excel Skills Job Simulation Virtual Experience Program
 - **Financial Statement**Entered the company's most recent financial statement.
 - **Capitalization Table & Overview:**Prepared a capitalization table, company, and industry overview page.
 - **Deal Structure & Forecast:**Developed a deal structure overview with quantitative & qualitative analyses & a 10-year financial model forecast.

SKILLS

- ***Software:** Matlab, Aspen Plus, Ansys
- ***Microsoft:** MS Excel, MS Powerpoint, MS Word
- ***Non Technical:** Project Management, Product Management
- ***Interpersonal:** Leadership, Communication, Problem Solving, Decision Making
- ***Language:** English, Hindi, Bengali

KEY COURSES TAKEN

- ***Mathematics:** Linear Algebra, basic Calculus, Complex analysis, Laplace , Probability & Random Processes
- ***Project Management:** Project Management 3.0
- ***Psychology of Learning:** Self Monitor, Reflex Thinking, constructive, sustainability & transformative learning
- ***Product Management:** Product Management 2.0
- ***Education for Sustainable Development:** SDGs, Circular economy, renewable energy & Sustainable agriculture
- ***Innovation by Design:** 7C's , Pitfalls of 7C's, Sunflower Solution of Pitfalls , 7R's

POSITION OF RESPONSIBILITY

- * **EVENT HEAD**, Indian Institute of Chemical Engineers, (IICChE), BIT Mesra Students Chapter *20 May, 2023 - Present*
 - Successfully Conducted Poster Presentation event which had a foot fall of 50+ audience
 - Also responsible for the content initiatives of the Club across platforms

ACHIEVEMENTS

- * **JEE MAINS**, Secured a rank of 52,477, placing in the top 5% of 10,00,000 students across India. *2021*
- * **WBJEE**, Secured 2000 placing in the top 2% of 1,00,000 students across India *2021*